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NIHR Invention for Innovation Programme

Scheme	i4i Product Development Awards
Funding Opportunity	2025/405 i4i Product Development Awards Researcher-Led

SUMMARY INFORMATION

Application Title	AI-Enabled Digital Sleep Diary: Clinical Validation, Usability and NHS Real-World Evaluation
Contracting Organisation	Sleep and Health Clinic App Limited
Contracting Organisation Department	Research & Development
Duration (months)	24
Total Cost to NIHR	£1,523,473.48
Has this application been previously submitted to this or any other funding body?	No

LEAD APPLICANT & RESEARCH TEAM

Lead Applicant

Full Name	Dr Venkata McNeillis
Organisation	Sleep and Health Clinic Ltd
Department	Sleep Unit
Proposed Role	The lead applicant will manage and ensure that the project meets the development timelines, work with the project team to design the research methodology, lead and coordinate the research team at UCLH. She will be the liason for the grant and ensure fiscal management is within the grant guideline.
ORCID	0009-0004-7611-2639
% FTE Commitment	100.00%

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Co-Applicants

Full Name	Miss Sasha McNeillis
Organisation	Sleep and Health Clinic Ltd
Department	Business Development/Innovation
Proposed Role	I will coordinate project activities, maintain timelines and milestones, support communication across partners, manage documentation, and assist with reporting, governance, and administrative requirements to ensure the study is delivered efficiently and to NIHR standards.
ORCID	0009-0003-1588-4750
% FTE Commitment	100.00%

Full Name	Mr Maxime Sibille
Organisation	Sleep and Health Clinic Ltd
Department	Management
Proposed Role	Deputy CEO responsible for product development and leading the platform to market readiness.
ORCID	0009-0001-9205-4676
% FTE Commitment	100.00%

Full Name	Mr Terence Hook
Organisation	Sleep and Health Clinic Ltd
Department	Technical Development
Proposed Role	I will oversee all technical development, ensuring the app, backend and analytics engine meet required performance, security and interoperability standards. I will coordinate engineers, manage technical risks, deliver updates and support DTAC and implementation requirements.
ORCID	0009-0002-0679-3876
% FTE Commitment	100.00%

Full Name	Miss Maria Bozzini
Organisation	UNIVERSITY COLLEGE LONDON HOSPITALS CHARITIES
Department	Sleep Unit
Proposed Role	To manage and ensure the integrity of research data, assisting with preliminary analysis and literature reviews, and supporting the drafting of reports and manuscripts.
ORCID	0000-0003-2972-3934
% FTE Commitment	40.00%

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ADMINISTRATIVE INFORMATION

Additional partners and/or collaborators

Organisations	

APPLICATION DETAILS

Plain English Summary

The problem

Sleep problems affect a third of adults in the UK, yet many people with sleep disorders including insomnia and sleep apnoea do not receive a timely or accurate diagnosis. Long waiting lists, limited specialist training in sleep medicine, and the use of handwritten sleep diaries cause delays and unnecessary referrals. These challenges place pressure on sleep clinics, GPs and other NHS services. Untreated sleep problems also contribute to long-term health conditions such as high blood pressure, diabetes, dental issues and mental-health difficulties.

The innovation

Sleep and Health App Limited (SHA) is a digital sleep-health app that helps people record their sleep patterns over 14 days. Instead of using a paper diary, patients enter simple information into the app each day. The app uses Artificial Intelligence (AI) to spot patterns that may indicate insomnia, disrupted sleep cycles or possible sleep-apnoea risk. It then produces a clear report that GPs, pharmacists and sleep specialists can use to support diagnostic and early triage. The aim is to make assessments faster, more consistent and seamless for both patients and clinicians.

What this project will do

Over 24 months, this project will develop, test and prepare SHA for NHS use. The work includes:

• A clinical validation study at University College London Hospitals (UCLH) comparing SHA reports with clinician-reviewed handwritten sleep diaries.

• Usability and accessibility testing with people from a wide range of backgrounds, including older adults, people with lower digital confidence and those from minority ethnic communities.

• A real-world evaluation in GP and community settings to understand how SHA affects consultation time, diary completion, referral accuracy and the patient experience.

• Health-economic analysis to confirm the potential of the app to streamline the overall sleep pathway, including reducing waiting time and unnecessary secondary care referrals, and cost savings.

• Completion of NHS requirements including DTAC (Digital Technology Assessment Criteria) and the NICE (National Institute for Health and Care Excellence) Evidence Standards Framework.

All data will be stored securely and handled according to NHS information-governance standards.

Working with patients and the public

People with lived experience of sleep problems helped shape the design of SHA. They told us that paper diaries are difficult to complete and that clear visual summaries and reminders would make the process easier. A Patient Advisory Group will continue to guide the project, helping ensure the app is simple to use, accessible and suitable for people with disabilities as well as different cultural and literacy needs.

Expected impact

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If successful, SHA will provide the NHS with a quick, easy-to-use digital tool that improves early sleep assessment. It could:
• Improve early triage of patients with sleep disorder symptoms
• Shorten patient pathway and waiting times
• Alleviate system pressure by accelerating the diagnostic stage
• Improve early detection of related health conditions
• Reduce consultation length for clinicians
• Reduce unnecessary secondary care referrals
• Reduce pressure on mental-health services and long-term condition pathways

By replacing paper diaries with a validated digital tool, this project aims to modernise the sleep pathway, improve access to timely care and support better health outcomes for people experiencing sleep difficulties.

ACKNOWLEDGEMENT AND CONFLICTS

I have read and understood the terms of applying as Lead Applicant and accept this role	✓

Potential Conflicts	None

Research Plan – NIHR i4i Product Development Award

Lead Applicant (SME path): Sleep and Health App Ltd (“SHA”)

Project Duration: 24 months

1. Background and Rationale

Sleep disorders are common, underdiagnosed and often mismanaged across the UK. Around one-third of adults' report sleep problems, 29% experience insomnia symptoms, and up to 85% of the 1.5 million people with obstructive sleep apnoea remain undiagnosed. Limited clinician training, diary variability and long waiting lists delay diagnosis, drive inappropriate referrals and increase system burden.

Current NHS pathways rely on paper sleep diaries, which are labour-intensive, inconsistently completed and difficult to interpret. Commercial sleep apps exist, but none provide validated diagnostic reports for NHS use. Gold-standard tools such as actigraphy and polysomnography (PSG) are expensive, capacity-limited and unsuitable for early triage at population scale. This creates a clear unmet need for a validated, accessible digital diagnostic tool that can support early assessment, NHS decision-making and has the potential to reduce costs to the NHS.

SHA is a mobile-based AI-enabled digital diagnostic tool delivered via a smartphone app (iOS and Android), designed to replace handwritten sleep diaries with automated, standardised analysis. A TRL 6 prototype has shown promising usability and alignment with clinician interpretation of sleep-diary data.

Pathway modelling projects that SHA could deliver meaningful system-level efficiencies, including:

- Reducing consultation episodes by ~50%
- Reducing unnecessary specialist referrals by >50%
- Shortening referral-to-treatment timelines by ~50%
- Supporting prevention across multiple long-term condition pathways
- Reducing workforce pressure

These figures represent modelled projections based on current pathway assumptions. The PDA will formally test, validate and refine these estimates through the clinical validation (WP2), real-world evaluation (WP4) and health-economic modelling (WP5).

Despite these promising projections, two core hypotheses now require robust empirical validation at scale:
(1) SHA performs comparably to existing assessment tools (handwritten diary, and where feasible actigraphy/PSG), and
(2) SHA reduces consultation burden, unnecessary referrals and NHS costs.

It will generate the clinical validation, usability, real-world service-impact data and health-economic modelling required for adoption. It will also deliver the evidence required for DTAC, NICE ESF and NHS adoption.

2. Aims and Objectives

2.1 Overall aim

The overall aim is to optimise, evaluate and prepare SHA for NHS adoption as an AI-enabled digital diagnostic tool supporting early sleep-disorder triage and prevention of multimorbidity. We will use patient public participation within this development to shape the user interface and user experience of the product to enable it to meet the needs of relevant and underserved groups.

This project will explicitly evaluate the two core hypotheses (clinical performance and NHS cost/efficiency impact) while generating the regulatory, usability, inclusivity and real-world evidence needed for commissioners, ICSs and NICE-aligned decision-makers.

2.2 Objectives Table

Objective	*Description*
O1	Refine AI algorithm, reporting engine and interface to achieve clinical-grade reliability and usability.
O2	Conduct usability & accessibility testing with diverse and underserved patient groups.
O3	Undertake a pragmatic real-world evaluation in GP/community settings to assess acceptability, workflow fit, utility and system impact. With active engagement of patients, end-users and public.
O4	Conduct health-economic modelling demonstrating cost savings, reduced consultations and avoided referrals.
O5	Develop regulatory compliance to meet DTAC + NICE ESF requirements.
O6	Produce an NHS adoption and implementation strategy, including ICS-ready materials and commissioner evidence packs.

3. Methodology / Plan

3.1 Project Design and Methods

SHA uses a rule-based machine-learning system trained on anonymised patient-reported time-series data. The system analyses temporal patterns, extracts key features (sleep regularity, awakenings, efficiency), and flags insomnia, circadian irregularity and potential sleep-apnoea risk. It is explainable and aligned with the 2024 MHRA AI regulatory strategy.

This project will refine the algorithm, validate SHA against clinician-reviewed handwritten diaries (and actigraphy/PSG where available), and assess fairness across age, sex, gender, ethnicity and deprivation in line with NIHR inclusive-research requirements. The validation study will quantify diagnostic concordance, false-positive/negative patterns, safety and clinical usability. Real-world evaluation will assess consultation time, referral appropriateness, workflow integration and patient experience.

A health-economic analysis will estimate cost savings, workforce impact, avoided referrals and reduced waiting times, feeding directly into DTAC/NICE ESF evidence generation and commissioner-ready adoption materials. The app is free for users.

Primary endpoint: concordance between SHA outputs and clinician-reviewed diaries (and actigraphy/PSG where available). Secondary endpoints: usability, diary completion, consultation time, referral appropriateness, patient experience and projected cost-effectiveness.

3.2 Work Packages

Work Package	*Timeline (Months)*	*Description / Key Activities*
WP1 – Proof of Concept (POC) Optimisation	**1–2**	• Algorithm refinement (logic, features, risk rules) • UI/UX adjustments • Cybersecurity updates • Initial DTAC mapping
WP2 – Clinical Validation Study (UCLH)	**1–7**	• Independent NHS-led validation • Comparative assessment vs handwritten diaries • Diagnostic concordance & safety monitoring • PPI input into protocol & materials • Validation summary informing WP3 & WP4
WP3 – Usability & Accessibility Testing	**1–15**	• Recruit diverse user groups (sex, gender, ethnicity, deprivation, digital literacy) • Think-aloud sessions & SUS scoring (≥80) • Accessibility refinements (text size, contrast, readability) • Iterative UI/UX updates based on WP2 insights
WP4 – Real-World Evaluation (Pragmatic NHS Deployment)	**14–23**	• Deployment in GP & community settings • Comparator: handwritten diary pathway • Outcomes: acceptability, reach, fidelity, sustainability • Workflow observations & consultation-time data • Referral appropriateness assessment • Qualitative interviews & thematic analysis (MRC framework)
WP5 – Health Economics, NICE ESF & DTAC Compliance	**22–24**	• Cost-effectiveness & budget-impact modelling • Workforce & downstream system burden analysis • NICE ESF evidence pack (EfE, Ei, Safety, Usability) • DTAC compliance (security, interoperability, data protection, CSF) • Commissioner business case

3.3 Timeline & Milestones

- Month 2: WP1 POC optimisation complete
- Month 7: WP2 UCLH clinical validation study completed

- Month 12: Interim usability dataset (WP2)
- Month 15: Usability testing complete
- Month 18: Interim real-world evaluation dataset (WP4)
- Month 23: Real-world evaluation complete
- Month 24: Health-economic model + NICE ESF + DTAC pack delivered

3.4 Study Management

- Programme Lead: Maxime Sibille
- Technical Lead: Terry Hook
- AI/Data Lead: Brian Healy
- Clinical Lead: Dr Sara McNeillis
- Research Assistant: Maria Bozzini
- Project Officer: Sasha McNeillis
- Subcontractor: aSource Global (UK software house)

A quarterly Steering Committee will include:

- ICS representative
- Health economist
- Implementation-science expert
- Patient representative

3.5 Working with People & Communities (PPIE)

A Patient Advisory Group (8–10 members) will guide participant materials, usability testing, diagnostic-report clarity and accessibility improvements. Recruitment will deliberately include underserved groups: older adults, minority ethnic communities, people with low digital literacy and those experiencing socioeconomic disadvantage.

In line with NIHR's 2025 inclusive-research and sex/gender requirements, sampling will capture variation across age, sex, gender identity, ethnicity, deprivation and digital literacy**.** Outcomes will be analysed using demographic disaggregation where feasible, ensuring equitable performance and avoiding algorithmic bias. All materials will be ≤8th-grade reading age, and the app includes screen-reader, high-contrast and large-text modes.

3.6 Knowledge Mobilisation, Dissemination and Impact

The project will deliver:

- an ICS-ready implementation guide
- a NICE ESF evidence summary (EfE, Ei, safety, usability)
- a DTAC compliance pack
- a commissioner business case
- an MRC-aligned process-evaluation report

Evidence from WP2–WP4 will be shared with HINs, ICS boards and commissioners, supporting NHS adoption and commissioning decisions.

3.7 Research Team

The multidisciplinary team covers:

- Sleep medicine
- AI/data science
- Implementation science
- Health economics
- Commercialisation
- PPIE and end-user engagement

This meets PDA expectations for SME-led multi-applicant teams.

3.8 Inclusive Research

The study meets NIHR's updated inclusive-research and sex/gender requirements. Inclusion, data collection, analysis and knowledge mobilisation will explicitly consider representation across age, sex, gender, ethnicity, deprivation and digital literacy. Statistical analysis will include demographic disaggregation where appropriate. All exclusion criteria will be evidence-based and justified.

3.9 Intellectual Property & Commercialisation

Background IP includes the SHA algorithm, analytics engine, UX and reporting templates, owned by SHA Ltd. Foreground IP arising from optimisation and validation will be retained by SHA Ltd, with collaborator IP managed through formal agreements (UCLH for validation; aSource for software development).

A preliminary freedom-to-operate assessment has identified no blocking patents in digital sleep-diary interpretation, risk-flagging logic or reporting structures. A full patentability review will be conducted in Months 18–22.

Commercial pathways include NHS Supply Chain, G-Cloud, NHS App Library (post-ORCHA), and ICS-level licensing. Economic evidence from WP5 will support commissioning decisions and potential MedTech Funding Mandate positioning. Global opportunities include primary care, pharmacy-based sleep pathways, employer health and international health-system licensing. SHA uses an NHS-facing licensing model, generating revenue through contracts with providers or ICSs. There is no direct charge to patients, and long-term sustainability will come from NHS commissioning and digital procurement frameworks.

4. Key Performance Indicators (KPIs)

KPI	Target
SUS usability score	≥80
Referral-to-diagnosis reduction	≥50%
Unnecessary specialist referrals reduction	≥53%
Patient diary completion	≥85% (14 days)
Clinician acceptance	≥80% positive
Demographic fairness	≤10% variance across sex/gender/ethnicity

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Work Package Timeline for SHA Development and Evaluation

